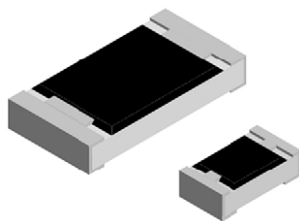


Thick Film Surface Mount Chip Resistors, Wraparound, Extremely Low Value (0.01 Ω to 0.976 Ω)



LINKS TO ADDITIONAL RESOURCES



FEATURES

- Extremely low resistance values (0.01 Ω to 0.976 Ω)
- Sulfur resistant (per ASTM B809-95 humid vapor test)
- Enhanced power rating due to long side terminal construction (0612, 1020 types)
- Suitable for current sensing and shunts
- Metal glaze on high quality ceramic
- Protective overglaze
- Lead (Pb)-free solder contacts on Ni barrier layer
- AEC-Q200 qualified
- Material categorization: for definitions of compliance please see www.vishay.com/doc?99912



RoHS
COMPLIANT
HALOGEN
FREE

STANDARD ELECTRICAL SPECIFICATIONS

GLOBAL MODEL	CASE SIZE	POWER RATING $P_{70^{\circ}\text{C}}$ W	TEMPERATURE COEFFICIENT + ppm/ $^{\circ}\text{C}$	RESISTANCE RANGE Ω	TOLERANCE $\pm$ %	E-SERIES ⁽²⁾
RCWE0402 ⁽³⁾⁽⁴⁾	0402	0.125	400	0.033 to 0.05	5.0	24
			200	0.051 to 0.196	1.0, 5.0	24; 96
			100	0.2 to 0.976	0.5 ⁽¹⁾ , 1.0, 5.0	
RCWE0603 ⁽⁴⁾	0603	0.2	700	0.010 to 0.018	5.0	24
			400	0.02 to 0.0324	1.0, 5.0	24; 96
			200	0.033 to 0.105	1.0, 5.0	
			100	0.11 to 0.976	0.5 ⁽¹⁾ , 1.0, 5.0	
RCWE0805 ⁽⁴⁾	0805	0.25	400	0.010 to 0.018	5.0	24
			300	0.02 to 0.0324	1.0, 5.0	24; 96
			200	0.033 to 0.05	1.0, 5.0	
			100	0.051 to 0.976	0.5 ⁽¹⁾ , 1.0, 5.0	
RCWE0612 ⁽⁴⁾	0612	1.0	300	0.010 to 0.016	2.0, 5.0	24
			200	0.018 to 0.2	2.0, 5.0	24; 96
			100	0.205 to 0.976	1.0, 5.0	
RCWE1206 ⁽⁴⁾	1206	0.5	600	0.010 to 0.018	5.0	24
			300	0.02 to 0.0324	1.0, 5.0	24; 96
			200	0.033 to 0.05	1.0, 5.0	
			100	0.051 to 0.976	0.5 ⁽¹⁾ , 1.0, 5.0	
RCWE1210 ⁽⁴⁾	1210	1.0	500	0.010 to 0.018	5.0	24
			300	0.02 to 0.0324	1.0, 5.0	24; 96
			200	0.033 to 0.05	1.0, 5.0	
			100	0.051 to 0.976	0.5 ⁽¹⁾ , 1.0, 5.0	
RCWE1020 ⁽⁴⁾	1020	2.0	200	0.010 to 0.016	2.0, 5.0	24
			100	0.0162 to 0.976	1.0, 5.0	24; 96
RCWE2010 ⁽⁴⁾	2010	1.0	600	0.010 to 0.018	5.0	24
			300	0.02 to 0.0324	1.0, 5.0	24; 96
			200	0.033 to 0.05	1.0, 5.0	
			100	0.051 to 0.976	0.5 ⁽¹⁾ , 1.0, 5.0	
RCWE2512 ⁽⁴⁾	2512	2.0	600	0.010 to 0.018	5.0	24
			300	0.02 to 0.0324	1.0, 5.0	24; 96
			200	0.033 to 0.05	1.0, 5.0	
			100	0.051 to 0.976	0.5 ⁽¹⁾ , 1.0, 5.0	

Notes

- Power rating depends on the max. temperature at the solder point, the component placement density and the substrate material
- Part marking: Reference "Surface Mount Resistor Marking" (www.vishay.com/doc?20020)
- Temperature range of TCR rating is 0 $^{\circ}\text{C}$ to 150 $^{\circ}\text{C}$. TCR values are (+) range only with no (-) range values; 1/2 of previous tolerance range
- (1) Tight tolerance of 0.5 % is available for resistance values above 0.300 Ω (0402 size) and above 0.200 Ω (0603 to 2512 sizes)
- (2) Use E24 decades only for 5.0 % tolerance. E24 or E96 decades are available for 0.5 % and 1.0 % tolerance. Refer to standard decade table (www.vishay.com/doc?31001)
- (3) Terminal strength tested per AEC-Q200-006 with the exception of 0.75 kg force is used
- (4) Qualified to AEC-Q200 rev. D

GLOBAL PART NUMBER INFORMATION

Global Part Numbering Example: RCWE060351L0FNEA (visit www.vishay.net Vishay Dale parts numbering manual for all options)

R	C	W	E	0	6	0	3	5	1	L	0	F	N	E	A		
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GLOBAL MODEL
(8 digits)

RCWE0402
RCWE0603
RCWE0805
RCWE0612
RCWE1206
RCWE1210
RCWE1020
RCWE2010
RCWE2512

VALUE
(4 digits)

L = mΩ *
R = decimal
10L0 = 0.01 Ω
R470 = 0.47 Ω

Note:
* Use "L" for resistance
values < 0.1 Ω

TOLERANCE
(1 digit)

D = ± 0.5 %
F = ± 1.0 %
G = ± 2.0 %
J = ± 5.0 %

TCR
(1 digit)

K = +100 ppm/°C
N = +200 ppm/°C
M = +300 ppm/°C
Q = +400 ppm/°C
P = +500 ppm/°C
T = +600 ppm/°C
G = +700 ppm/°C

PACKAGING
(2 digits)

EA = lead (Pb)-free,
tape/reel

SPECIAL
(up to 2 digits)

(dash number)
from 1 to 99 as
applicable

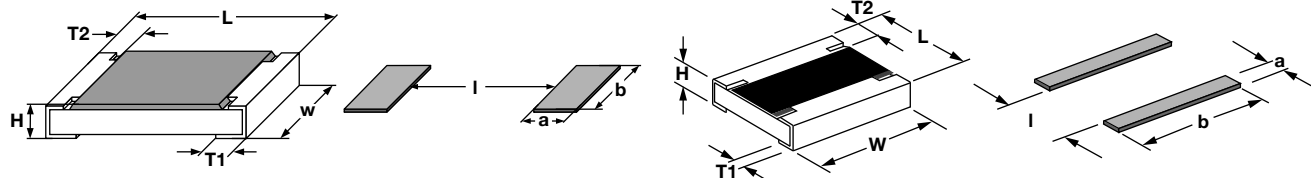
TECHNICAL SPECIFICATIONS

PARAMETER	UNIT	0402	0603	0805	0612	1206	1210	1020	2010	2512
Operating temperature range	°C	-55 to +155								
Maximum operating voltage	V	$(P \times R)^{1/2}$								
Insulation voltage U_{ins} (1 min)	V	> 75	> 100	> 200	> 100	> 300	> 300	> 300	> 300	> 300
Insulation resistance	Ω	> 10 ⁹								
Weight/1000 pieces (typical)	g	0.7	3	5.5	11.5	10.5	17.5	27.5	26	40.5

DIMENSIONS

RCWE0402 to RCWE2512

RCWE0612, RCWE1020

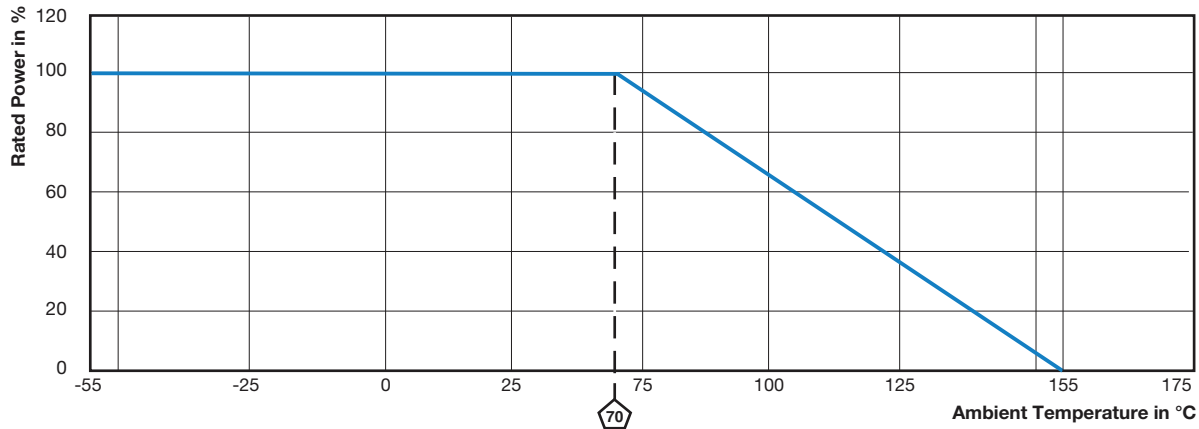


SIZE	DIMENSIONS in millimeters						SOLDER PAD DIMENSIONS in millimeters		
	RESISTANCE RANGE Ω	L	W	H	T1	T2	a	b	l
0402	0.033 to 0.976	1.05 ± 0.05	0.55 ± 0.05	0.35 ± 0.1	0.3 ± 0.15	0.25 ± 0.1	0.7	0.7	0.3
0603	0.01 to 0.03	1.6 ± 0.1	0.85 ± 0.1	0.5 ± 0.1	0.5 ± 0.2	0.3 ± 0.2	0.9	1.0	0.4
	0.033 to 0.976				0.3 ± 0.2		0.7	1.0	0.8
0805	0.01 to 0.03	2.0 ± 0.15	1.3 ± 0.1	0.55 ± 0.1	0.6 ± 0.2	0.35 ± 0.2	1.0	1.4	0.6
	0.033 to 0.976				0.4 ± 0.2		0.8	1.4	1.0
0612	0.01 to 0.976	1.6 ± 0.2	3.2 ± 0.2	0.6 ± 0.1	0.4 ± 0.15	0.25 ± 0.15	0.9	3.5	0.8
	0.01 to 0.03				0.9 ± 0.2		1.3	1.8	1.0
	0.033 to 0.05				0.8 ± 0.2		1.2	1.8	1.2
1206	0.051 to 0.976	3.1 ± 0.15	1.6 ± 0.15	0.6 ± 0.1	0.45 ± 0.2	0.45 ± 0.2	1.0	1.8	1.6
	0.01 to 0.03				0.8 ± 0.2		1.3	2.6	1.1
1210	0.033 to 0.976	3.1 ± 0.2	2.5 ± 0.2	0.6 ± 0.1	0.4 ± 0.2	0.4 ± 0.2	0.9	2.6	2.0
	0.01 to 0.03				0.8 ± 0.2		1.3	2.6	1.1
1020	0.01 to 0.976	2.5 ± 0.2	5.0 ± 0.2	0.6 ± 0.1	0.55 ± 0.15	0.30 ± 0.15	1.2	5.5	1.4
	0.01 to 0.03				1.6 ± 0.3		2.3	3.0	1.4
	0.033 to 0.05				0.7 ± 0.3		1.4	3.0	3.2
2010	0.051 to 0.976	5.0 ± 0.2	2.5 ± 0.15	0.6 ± 0.1	0.7 ± 0.3	0.6 ± 0.2	1.4	3.0	3.2
	0.01 to 0.03				2.0 ± 0.3		2.8	3.6	1.4
	0.033 to 0.05				0.8 ± 0.3		1.6	3.6	3.8
2512	0.01 to 0.03	6.3 ± 0.2	3.15 ± 0.15	0.6 ± 0.1	0.8 ± 0.3	0.6 ± 0.2	1.6	3.6	3.8
	0.051 to 0.976				0.8 ± 0.3		1.6	3.6	3.8

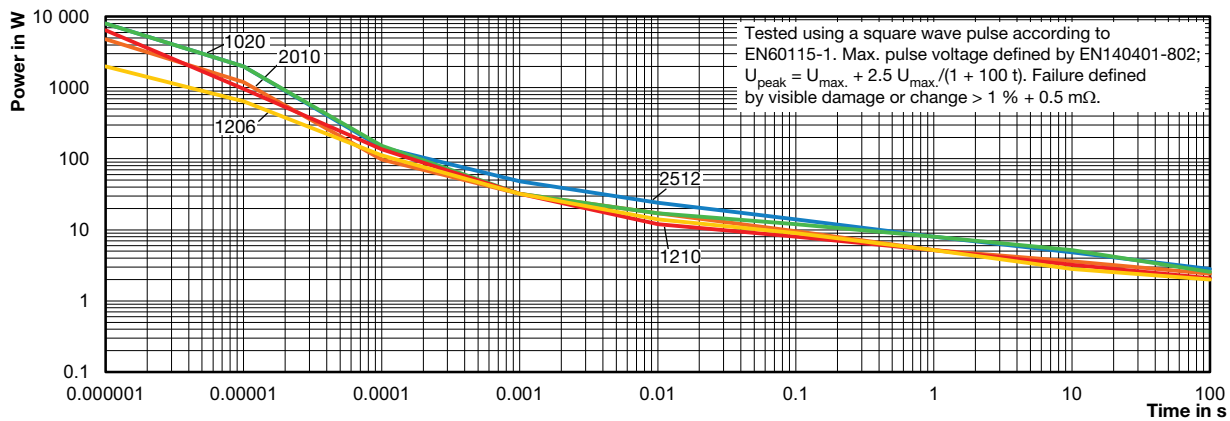
Notes

- 3D models available: www.vishay.com/doc?31106
- Surface mount solder profile recommendations: www.vishay.com/doc?31052

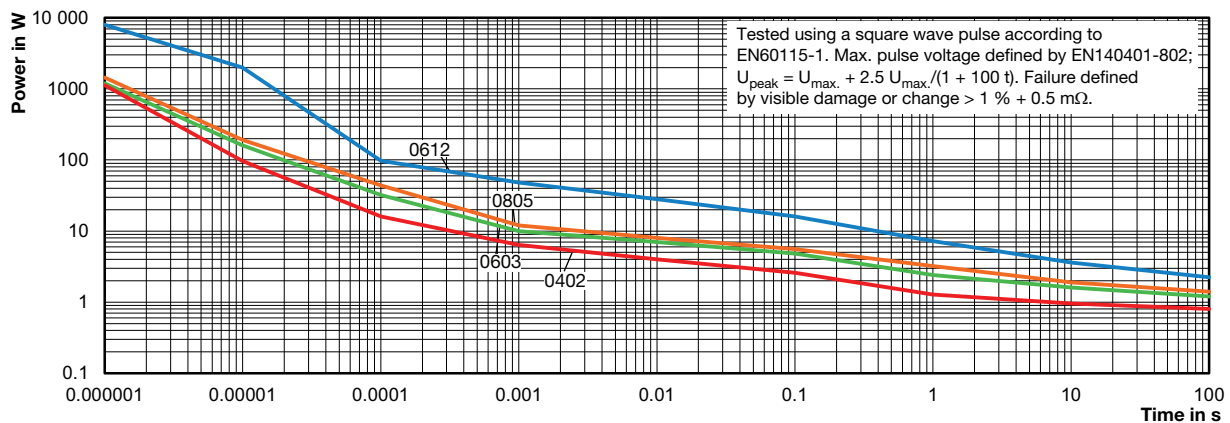
DERATING



SINGLE PULSE



SINGLE PULSE





PERFORMANCE		
TEST	CONDITIONS OF TEST	TEST LIMITS
Thermal shock	MIL-STD-202, method 107, -55 °C to +125 °C, 300 cycles at each extreme	± 1.0 % + 0.0005 Ω
Short time overload	2 x rated power; size and duration - 0402: 0.5 s, 0603 and 0805: 1 s, 1206 and larger: 2 s	± 0.5 % + 0.0005 Ω
High temperature exposure	MIL-STD-202, method 108, 1000 h at T = 125 °C, 0 % power	± 2.0 % + 0.0005 Ω
Temperature cycling	JESD 22, method JA-104, 1000 cycles (-55 °C to +125 °C)	± 2.0 % + 0.0005 Ω
Biased humidity	MIL-STD-202, method 103, 1000 h 85 °C / 85 % RH, 10 % x (P x R) ^{1/2}	± 2.0 % + 0.0005 Ω
Mechanical shock	MIL-STD-202, method 213, condition C, 10 g's, 6 ms (half sine), 3 directions	± 1.0 % + 0.0005 Ω
Vibration	MIL-STD-202, method 204, 5 g's, 20 min, 12 cycles, 3 directions, 10 Hz to 2000 Hz	± 1.0 % + 0.0005 Ω
Operational life	MIL-STD-202, method 108, 1000 h at T = 125 °C at rated power	± 2.0 % + 0.0005 Ω
Resistance to solder heat	MIL-STD-202, method 210, +260 °C solder, 10 s to 12 s dwell, 25 mm/s emergence	± 1.0 % + 0.0005 Ω
Moisture resistance	MIL-STD-202, method 106, 0 % power, 7a and 7b not required	± 2.0 % + 0.0005 Ω

Note

- Contact ww2bresistors@vishay.com for application specific performance requirements or qualification data. Typical performance is better than stated test limits

PACKAGING					
MODEL	REEL				
	TAPE WIDTH	DIAMETER	PITCH	PIECES/REEL	CODE
RCWE0402	8 mm / punched paper	180 mm / 7"	2 mm	10 000	EA
RCWE0603	8 mm / punched paper	180 mm / 7"	4 mm	5000	EA
RCWE0805	8 mm / punched paper	180 mm / 7"	4 mm	5000	EA
RCWE0612	8 mm / punched paper	180 mm / 7"	4 mm	5000	EA
RCWE1206	8 mm / punched paper	180 mm / 7"	4 mm	5000	EA
RCWE1210	8 mm / punched paper	180 mm / 7"	4 mm	5000	EA
RCWE1020	12 mm / embossed plastic	180 mm / 7"	4 mm	4000	EA
RCWE2010	12 mm / embossed plastic	180 mm / 7"	4 mm	4000	EA
RCWE2512	12 mm / embossed plastic	180 mm / 7"	8 mm	2000	EA

Notes

- Embossed carrier tape per EIA-481-1A
- Additional packaging details at: www.vishay.com/doc?31543

LINKS TO RELATED DOCUMENTS	
SELECTOR GUIDE	
Overview of Automotive Grade Products	www.vishay.com/doc?49924
TECHNICAL NOTES	
SMD Current Sense: AEC-Q200 vs. Vishay Qualification	www.vishay.com/doc?30416
MIL-PRF vs. AEC-Q200: Do You Know What You Are Getting?	www.vishay.com/doc?11000
WHITE PAPER	
Thermal Management for Surface-Mount Devices	www.vishay.com/doc?30380
Temperature Coefficient of Resistance for Current Sensing	www.vishay.com/doc?30405



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